

AN INTELLIGENT CROWD RISK MANAGEMENT SYSTEM

A MINI PROJECT REPORT

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ABSTRACT

The proposed system, Smart Crowd Monitoring and Alert System, is an intelligent surveillance solution designed to enhance safety in crowded environments such as temples, stadiums, public events, and transportation hubs. The system integrates real-time video processing and artificial intelligence to detect and manage crowd density effectively.

Using the object detection capabilities of YOLOv8, the system analyzes live video streams captured from multiple cameras, including mobile-based IP webcams. The backend, developed using Flask, processes the video feeds and counts the number of people in real time. Based on a predefined crowd threshold, the system determines whether the situation is normal or critical.

When the crowd exceeds the limit, the system automatically triggers alerts and sends notifications through Telegram using bot integration. This enables authorities to take quick action and prevent dangerous situations like stampedes.

The frontend dashboard provides an interactive interface displaying live camera feeds and crowd status. The system is scalable and can handle multiple camera inputs across different locations.

Overall, this project demonstrates the use of artificial intelligence for improving public safety, offering an efficient and practical solution for crowd management.

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol / Term	Description
AI	Artificial Intelligence
CV	Computer Vision
YOLO	You Only Look Once (Object Detection Algorithm)
API	Application Programming Interface
FPS	Frames Per Second
IP	Internet Protocol
CNN	Convolutional Neural Network
IoU	Intersection over Union
GPU	Graphics Processing Unit
NMS	Non-Maximum Suppression
RTSP	Real Time Streaming Protocol
HTTP	Hypertext Transfer Protocol